

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 897+866

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

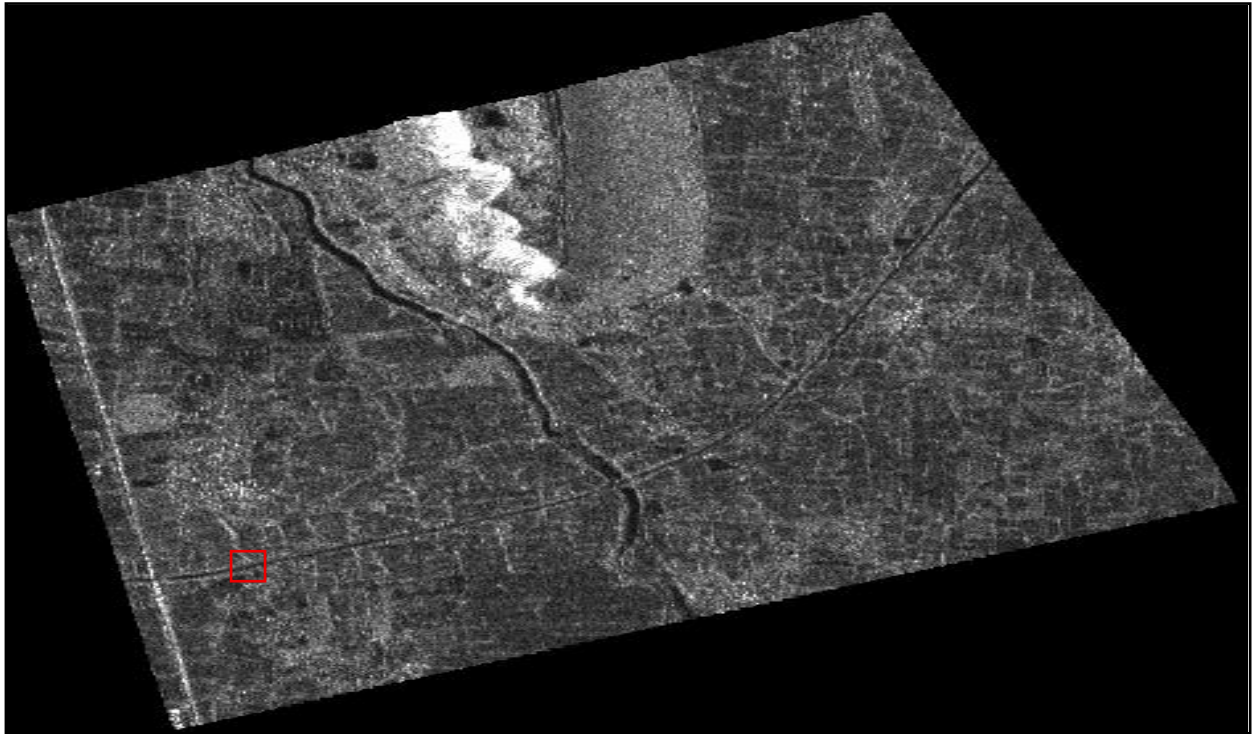
Flight Direction - DESCENDING

Polarization – VV+VH

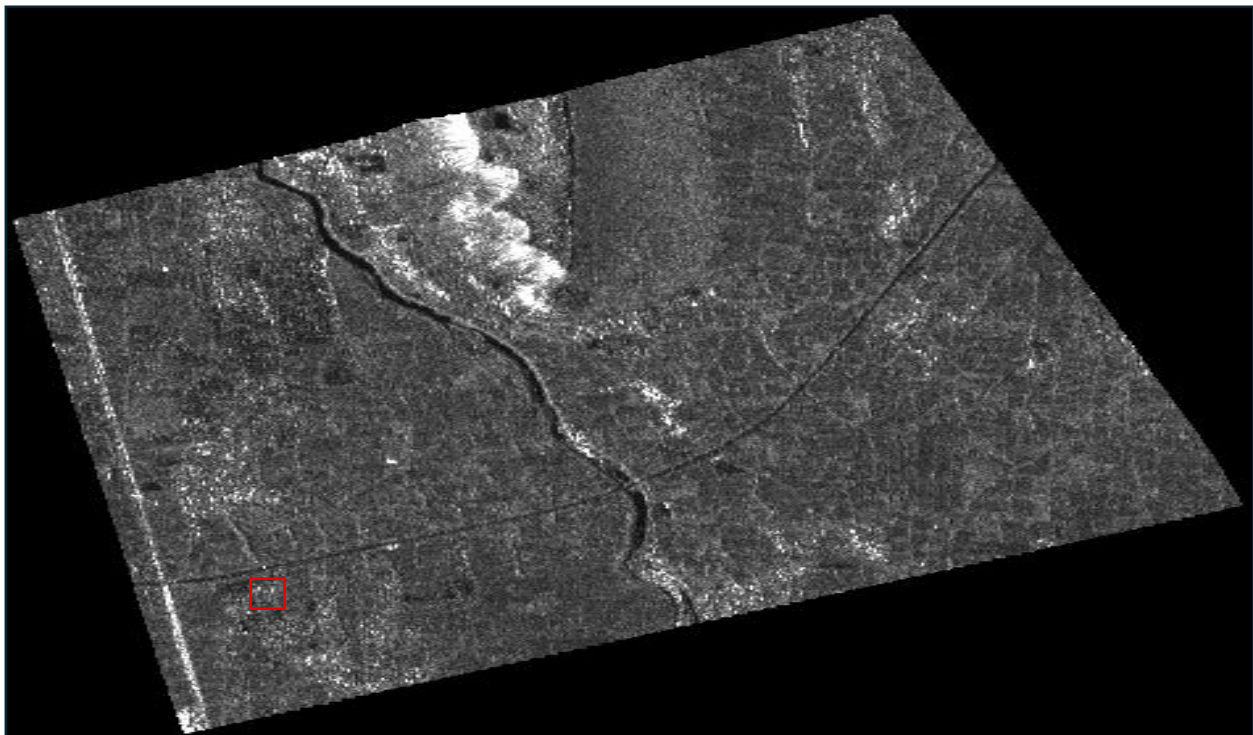
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

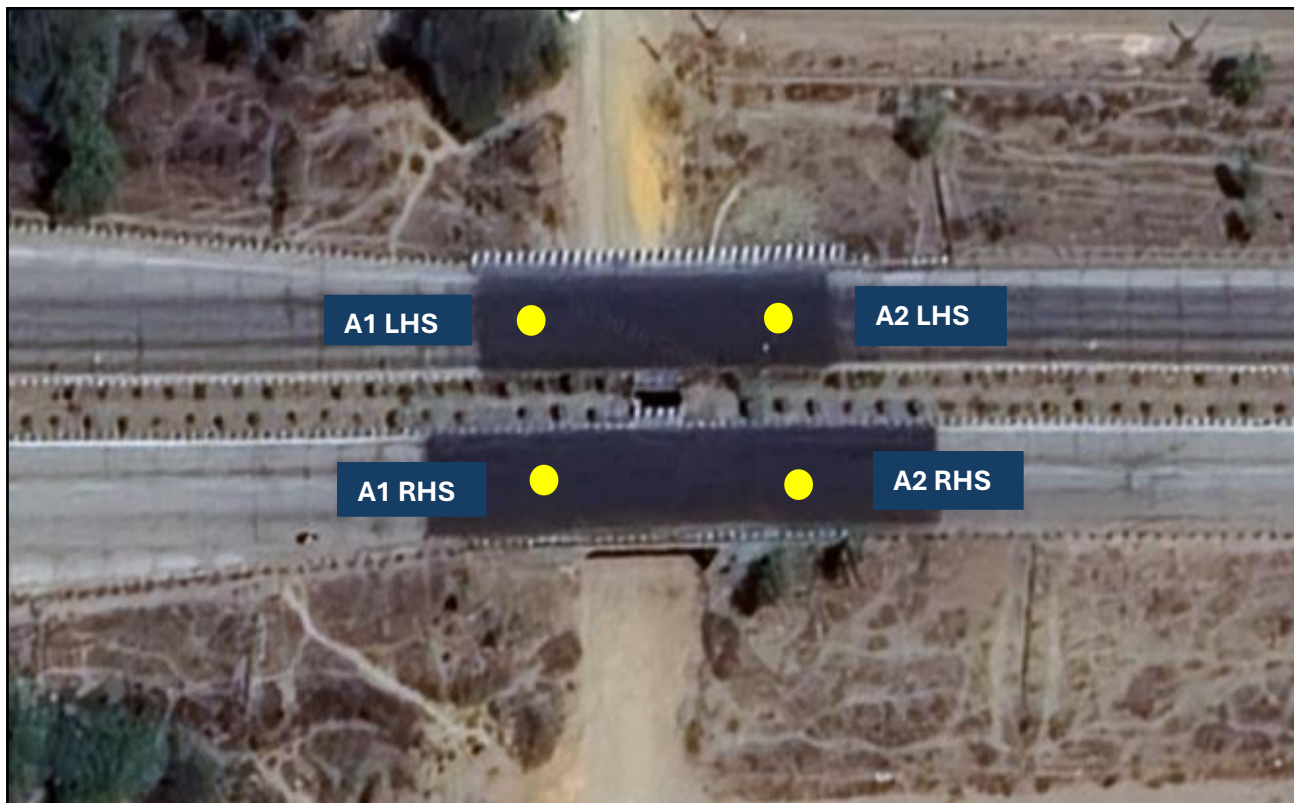


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

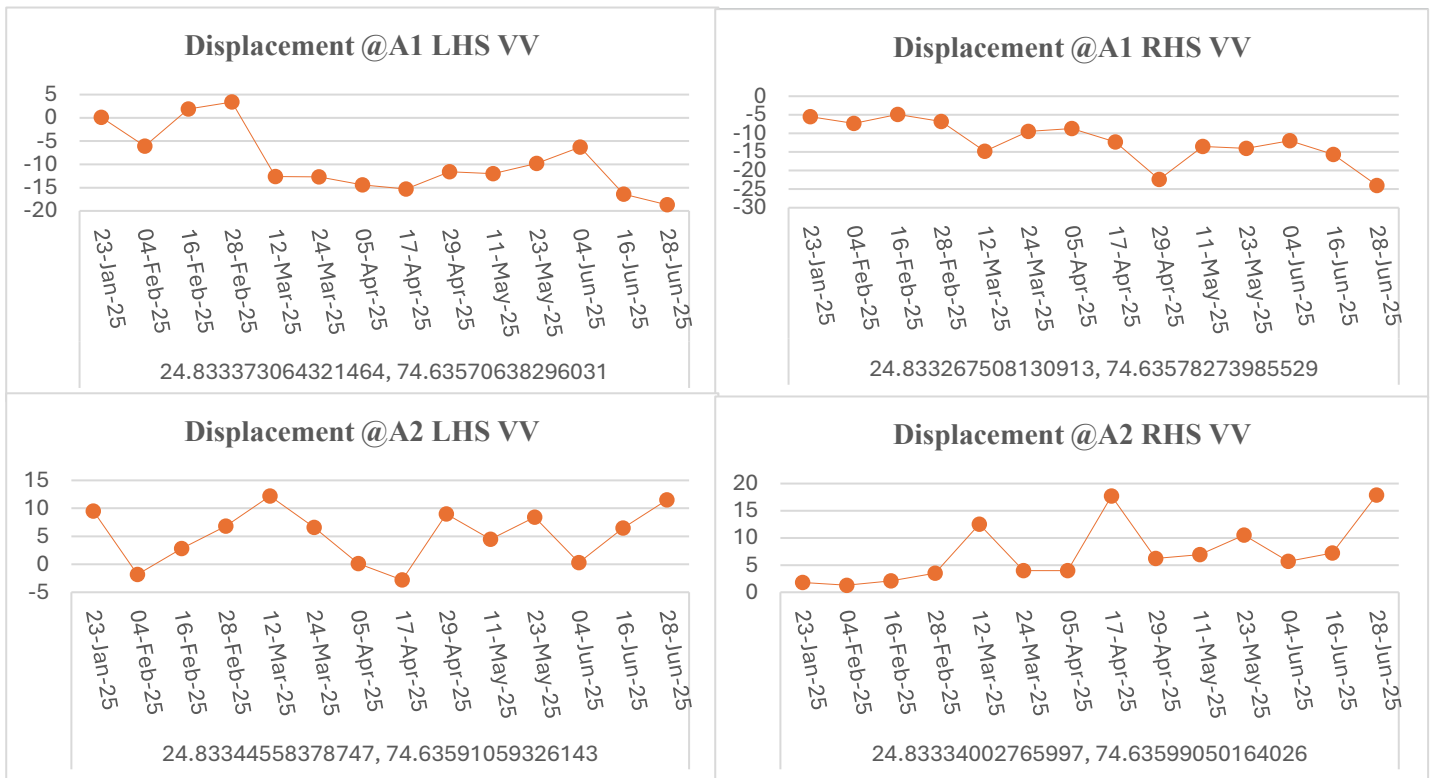
VUP – CH-897+866

Length – 32.02M

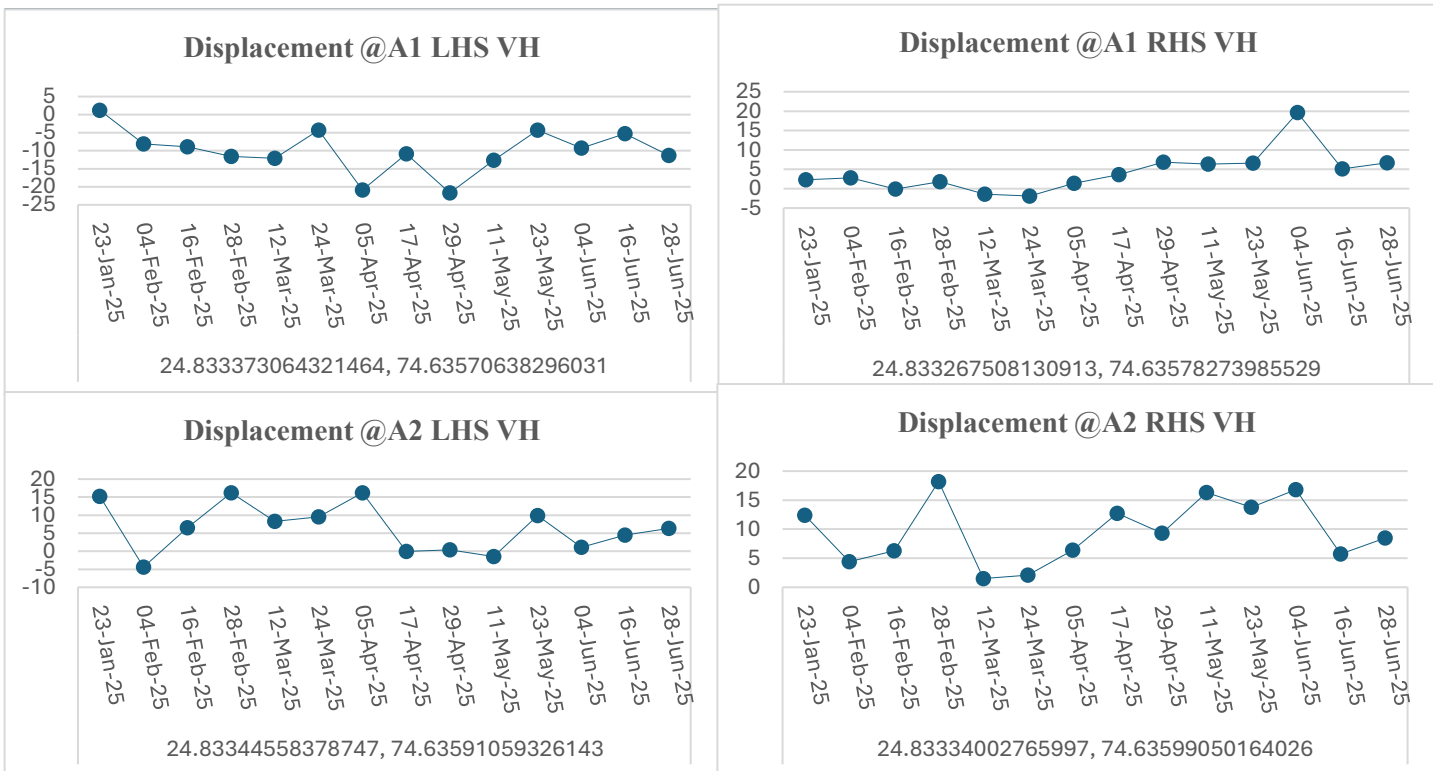


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



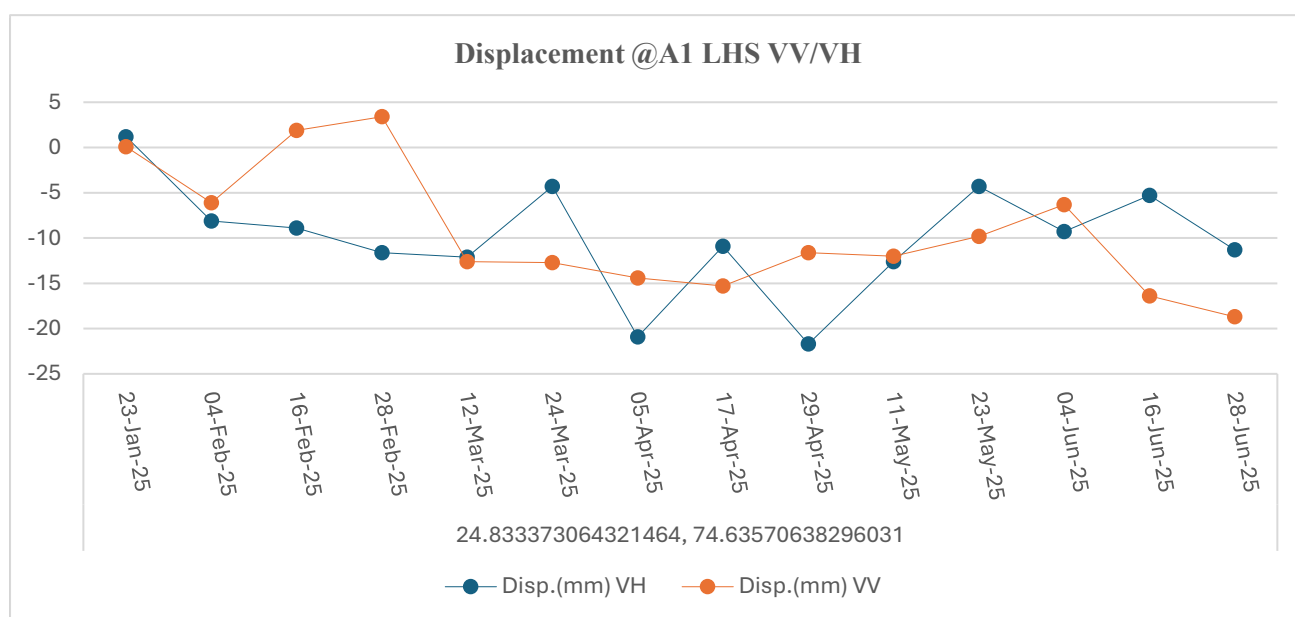
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.833373064321464, 74.63570638296031)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with an overall negative dominance over the monitoring period, indicating net vertical settlement at the A1 LHS location. VV values range approximately from +3 mm to –18 mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with persistent negative dominance, ranging roughly between +1 mm and –22 mm.

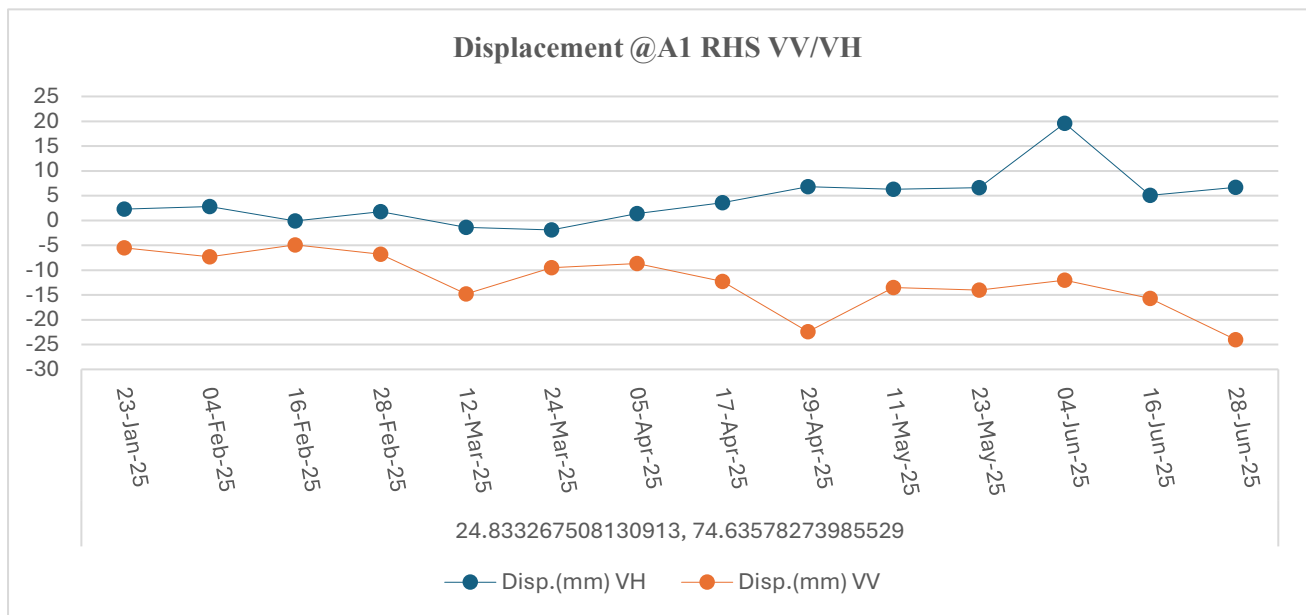
Critical Events & Anomalies:

- **February 2025:** VH continues declining to –9 mm, while VV remains mildly positive (+1 mm), indicating continued lateral settlement without vertical deterioration.
- **June 2025:** VV declines further to –16 to –18 mm, while VH fluctuates between –6 and –11 mm, confirming continued and progressive settlement toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.833267508130913, 74.63578273985529)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability with a predominantly positive trend, ranging roughly between 0 mm and +20 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits strong negative dominance with high variability throughout the monitoring period, indicating persistent vertical settlement at the A1 RHS location.

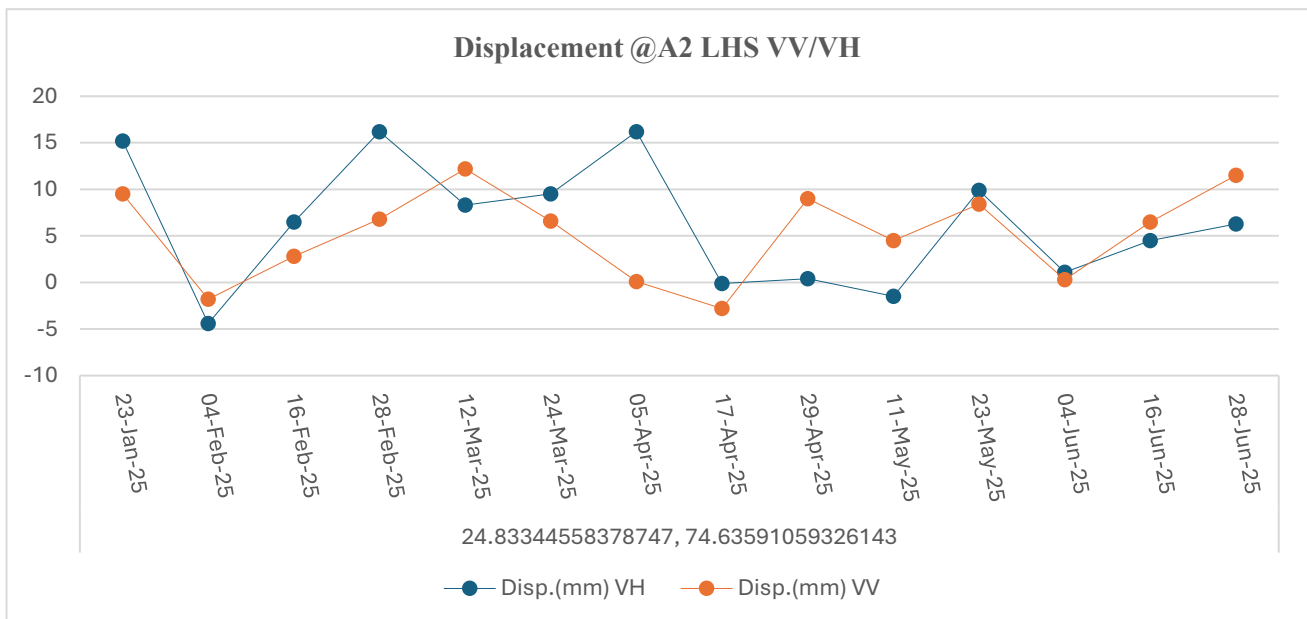
Critical Events & Anomalies:

- **January 2025:** VH begins near +2 mm, while VV is moderately negative (-5 mm), indicating early vertical settlement with limited lateral movement.
- **June 2025:** VH peaks near +20 mm, while VV remains around -13 mm, representing a strong lateral displacement pulse under continued vertical settlement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.83344558378747, 74.63591059326143)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows high variability with repeated oscillations, ranging roughly between -4 mm and $+16$ mm.
- **Vertical Displacement (VV)** exhibits moderate variability with alternating uplift and settlement phases over the monitoring period. VV values range approximately from -3 mm to $+12$ mm.

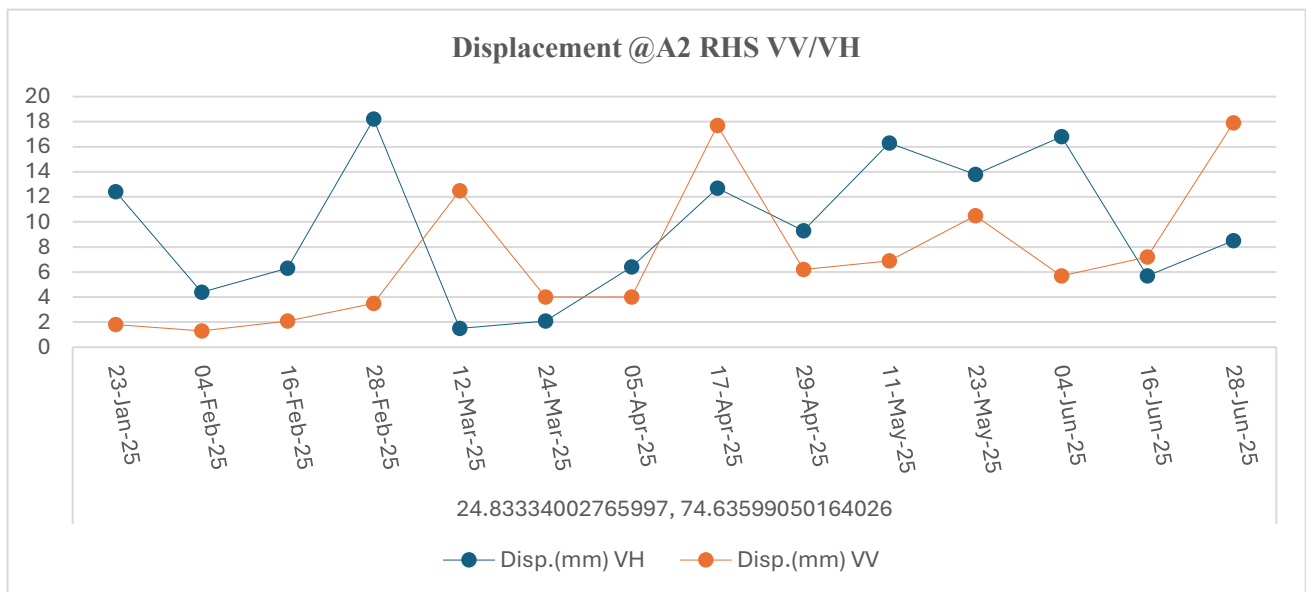
Critical Events & Anomalies:

- **January 2025:** VH begins high ($+15$ mm), while VV is also positive ($+9$ mm), indicating initial coupled lateral and vertical uplift.
- **February 2025:** Both VH and VV drop sharply, marking a short-term coupled settlement event.
- **June 2025:** VH increases to $+4$ – 6 mm, while VV rises to $+7$ – 11 mm, confirming renewed combined lateral and vertical uplift toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.83334002765997, 74.63599050164026)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits moderate variability with an overall positive dominance over the monitoring period, indicating net vertical uplift at the A2 RHS location.
- **Horizontal (VH) displacement** Horizontal displacement shows high variability with repeated uplift pulses, ranging roughly between 2 mm and 18 mm.

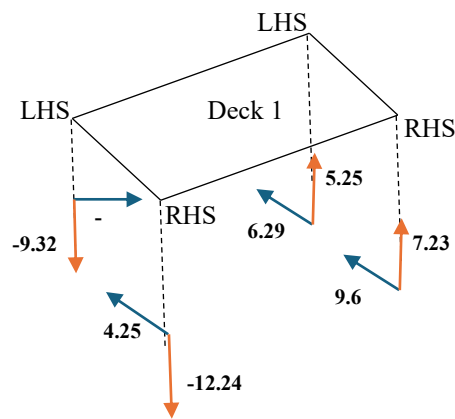
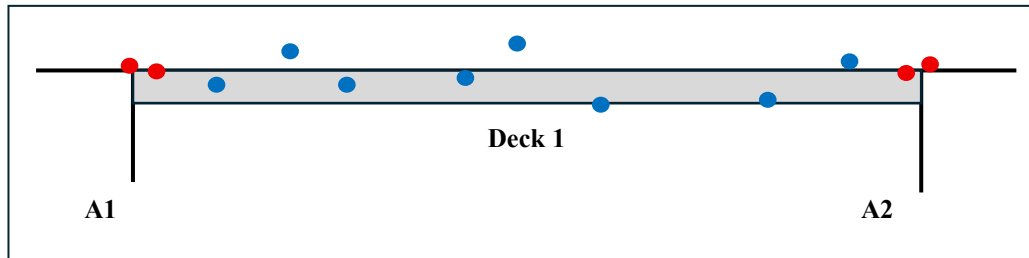
Critical Events & Anomalies:

- **February 2025:** VH spikes sharply to 18 mm, while VV remains low (4 mm), representing a lateral-dominant uplift pulse.
- **March 2025:** VH recovers to 6 mm, while VV reduces slightly to 4 mm, marking a short-lived stabilisation phase.
- **June 2025:** VV reaches 18 mm, while VH settles near 8 mm, confirming strong vertical uplift toward the end of the monitoring period.

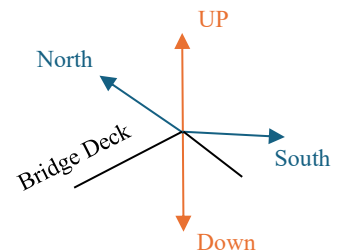
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinctly non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The simultaneous presence of uplift and settlement confirms vertical rotation combined with localized bending, rather than uniform vertical deflection.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates strong directional variability, clearly reflecting torsional deformation of the bridge deck slab. The opposing horizontal displacement directions strongly indicate torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab displays a torsion-dominated deformation response, characterized by significant lateral displacement combined with asymmetric vertical uplift and settlement. Overall, the observed displacement pattern is structurally significant, warranting continued monitoring and correlation with loading conditions, bearing behaviour, and environmental effects.